INCH-POUND

MIL-DTL-85097/7D(OS)
1 May 1996
SUPERSEDING
MIL-A-85097/7C(AS)
24 August 1984

ASSOCIATED DETAIL SPECIFICATION

AIRCREW ESCAPE PROPULSION SYSTEMS (AEPS); ROCKET MOTOR MARK 84 MODS 1 AND 2

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This associated specification covers requirements unique to the Rocket Motor Mk 84 Mods 1 and 2 for use in the S-3 Aircraft (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents specified in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

NAVAL AIR SYSTEMS COMMAND (CAGE Code 30003)

DL 503AS102	Rocket Motor Mark 84 Mod 1 (Seat Stabilizer)
DL 503AS200	Rocket Motor Mark 84 Mod 2 (Seat Stabilizer)

(Application for copies should be addressed to Commander, Naval Surface Warfare Center, Indian Head Division, 101 Strauss Avenue, Data Control Branch (Code 8410), Indian Head, MD 20640-5035.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Standardization Branch (Code 8420), 101 Strauss Avenue, Indian Head, MD 20640-5035 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by sending a letter.

AMSC N/A FSC 1377

DISTRIBUTION STATEMENT A: Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:23Z
Check the source to verify that this is the current version before use.

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications or specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The requirements in this associated specification, unless in direct conflict with MIL-A-85097, shall be considered additional, not substitute. The Mk 84 Mod 1 Rocket Motor shall conform to DL 503AS102 and the Mk 84 Mod 2 Rocket Motor shall conform to DL 503AS200. The complete requirements for acquiring the product described herein shall consist of this document and MIL-A-85097.

3.2 Preproduction.

3.2.1 Propellant aging. When tested in accordance with MIL-A-85097, the propellant shall meet the limit requirements specified by the contractor and which are acceptable to the Government.

3.3 Preproduction/production.

3.3.1 Propellant. The propellant shall conform to the contractor's in-house propellant specification when tested for data specified in MIL-A-85097. The contractor shall make available the results of the tested values and the in-house specification limits for comparison (see 6.2).

3.3.2 Ballistic requirements. The ballistic requirements shall be as specified in table I and provide the additional data as specified herein. See 6.4 for a definition of the parameters.

TABLE I. Ballistic performance requirements.

Parameter	-40 ± 5°F		160 ± 5°F	
	Min	Max	Min	Max
Max time from zero time to 70% peak thrust(s)	—	0.030	—	0.030
Peak thrust (lb)	550	980	550	980
Ignition peak thrust (lb)	—	—	—	1350
Burn time (s)	—	0.424	0.230	—
Action time (s)	—	0.480	0.278	—
Impulse (lb-s)	220	244	220	249
Torque about longitudinal axis (in-lb)	-6	+6	-6	+6

3.3.3 Unstable burning. Unstable burning does not apply to the Mk 84 Mod 1 and Mod 2 Rocket Motors.

3.3.4 Sear. The force required to withdraw the sear shall be not less than 25 pounds nor greater than 35

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:23Z
Check the source to verify that this is the current version before use.

pounds. Travel of sear to actuate the rocket motor shall be not less than 0.25 inch nor greater than 0.69 inch. Travel to completely remove the sear shall be 0.75 inch maximum.

3.3.5 Center of gravity. The calculated or measured center of gravity shall be within 0.03 inch of the axis of rotation.

4. VERIFICATION

4.1 Preproduction/production.

4.1.1 Conditioning temperatures. The conditioning temperatures and tolerance shall be as follows:

a. Cold: $-40 \pm 5^{\circ}F$
b. Hot: $160 \pm 5^{\circ}F$

4.1.2 Conditioning periods. Time for conditioning prior to static firing or during temperature cycling shall be as follows:

a. Static firing: 4 hours, minimum
b. Temperature cycling: 4 hours, minimum

4.1.3 Radiographic inspection. All components or assemblies shall be radiographed in two views 90 degrees apart.

4.1.4 Ballistic testing. Ballistic testing, for conformance to the requirements of 3.3.2, shall be performed as specified in MIL-A-85097.

4.1.5 Sear pull test. Sear pull tests shall be conducted on all rocket motors. The sear force shall be statically determined, before final assembly, by means of a pull test fixture. The sear pull velocity shall be 18 ± 2 ft/s. Test results shall be made available to the procuring activity with DCASR acceptance (see 6.2).

5. PACKAGING

(This section is not applicable to this associated specification.)

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The Mk 84 Mod 1 and Mod 2 Rocket Motors are used as a means of stabilizing the ejection seat during an emergency escape of personnel from the S-3A Aircraft.

6.2 Acquisition requirements. Acquisition documents should specify the following:

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:23Z
Check the source to verify that this is the current version before use.

a. Title, number, and date of this specification
b. Mark and Mod designation of rocket motor (see 3.1)
c. Format for records of test results (see 3.3.1, 4.1.5, and 6.3)

6.3 Test records. A guideline for the format of required records of test results is needed to ensure that all parameters are recorded. A Data Item Description for propellant analysis report for Aircrew Escape Propulsion Systems (AEPS) devices (see 3.3.1) and for test reports (see 4.1.5) will be adequate.

6.4 Definitions.

6.4.1 Time from zero time to 70 percent peak thrust. Time to 70 percent peak thrust is the interval between the ignition signal (sear pin release) until 70 percent of maximum thrust (the highest value of thrust) is achieved on the initial rise portion of the thrust-time curve.

6.4.2 Peak thrust. Peak thrust is the maximum value of thrust that occurs after the ignition peak thrust time interval.

6.4.3 Ignition peak thrust. Ignition peak thrust is the value of thrust that occurs within the first 0.030 second after the first indication of thrust.

6.4.4 Torque. Torque is measured during the rocket motor burn (time after the ignition peak thrust time interval).

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Preparing Activity:
Navy - OS
(Project 1377-0042)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:23Z
Check the source to verify that this is the current version before use.